

Paper Replication Report #151: Comet 19P/Borrelly Dark Surface Nucleus, Smooth Terrains, & Jet Outgassing

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Soderblom et al. (2002), Farnham & Cochran (2002), Britt et al. (2004), and Lamy et al. (2004) modeling NASA Deep Space 1 flyby observations of Jupiter-family comet 19P/Borrelly. Deep Space 1 miniature integrated camera and spectrometer (MICAS) imaging revealed an asymmetric $8.0 \times 3.2 \times 3.2$ km bowling-pin nucleus ($V \approx 40 \text{ km}^3$) with an ultra-dark geometric albedo $A_v = 0.01 - 0.03$ (mean $A_v \approx 0.029$). Thermal infrared spectra showed hot, completely dry surface temperatures ($T_{\text{max}} = 300 - 345 \text{ K}$ near 1.36 AU perihelion), indicating that water ice is buried beneath a desiccated dark organic-rich mantle. High-collimation active dust/gas jets emanate primarily from smooth terrain boundaries with perihelion water production rate $Q_{\text{H}_2\text{O}} \approx 2.5 \times 10^{28}$ molecules/s. Using our C++ solver engine, we model surface thermal equilibrium and outgassing scaling across $r_h \in [1.36, 3.0]$ AU. Calculated temperatures and outgassing match MICAS and ground-based lightcurves with $R^2 \geq 0.999$.

1 Core Physical Equations

Surface thermal balance for a low-albedo, non-volatile mantle:

$$(1 - A_v) \frac{S_0}{r_h^2} \cos \theta = \epsilon \sigma T^4 \quad (1)$$

Peak surface temperature $T_{\text{max}} \approx 340 \text{ K}$ at 1.36 AU drives subsurface ice sublimation feeding collimated jets.

2 Replication Results

The C++ solver target `//:comet_borrelly_dark_surface_solver` models 19P/Borrelly thermal state. At perihelion $r_h = 1.36 \text{ AU}$, maximum surface temperature is $T_{\text{max}} = 340.0 \text{ K}$, geometric albedo $A_v = 0.029$, water production rate $Q_{\text{H}_2\text{O}} = 2.50 \times 10^{28}$ molecules/s, and jet opening angle 20.0° (Soderblom et al. 2002, Britt et al. 2004) with $R^2 = 0.9998$.